Htrack26: Interim Project Progress Report

Group 5

Abstract—Htrack26 is an ongoing project to develop a multi-role digital health platform for patients, doctors, and researchers. At the current development stage, the team has completed the initial Flutter application, Supabase authentication and database connection, health-record management, and basic analytics. This report presents the work completed so far, the current limitations, and the activities planned for the next phase.

Index Terms—Digital Health, Flutter, Supabase, Health Records, Data Visualization, Project Progress.

I. PROJECT OBJECTIVE

Htrack26 is being developed as a shared health platform with three intended user groups. Patients will be able to maintain personal health records; doctors will be able to review information shared by their patients; and researchers will be able to work with appropriately anonymized health data. The current work focuses on building the application's core foundation.

II. WORK COMPLETED

The team has set up the Flutter/Dart application and connected it to Supabase. User registration and login, profile retrieval, and the main dashboard are working. The PostgreSQL database includes profiles and health_records tables, with role values for patient, doctor, and researcher.

III. HEALTH RECORDS

A manual record-entry interface has been developed for blood pressure, heart rate, oxygen saturation, temperature, weight, height, waist measurement, blood glucose, selected laboratory values, daily steps, sleep duration, and water intake. Records are saved to Supabase and can be viewed and deleted through the history screen.

IV. ANALYTICS PROGRESS

The analytics section now organizes measurements into Vital Signs, Body Measurements, Blood Glucose, Laboratory, and Lifestyle. Individual measurement screens display line charts and basic statistics such as average, minimum, maximum, and record count. The team has also integrated the fl_chart package for visualization.

V. CURRENT DEVELOPMENT STATUS

The patient-side record-entry and basic analytics workflow is the most developed part of the system. Doctor and researcher roles are currently defined in the database, but their dedicated dashboards, workflows, and access permissions have not yet been completed. Medical-report upload and OCR are also planned features, not completed features.

VI. NEXT PHASE

The next phase will focus on improving analytics for all measurement types, implementing role-based interfaces, and designing secure doctor–patient access. The team also plans to investigate medical report upload and OCR extraction, followed by anonymized research-data access and expanded testing.

VII. PROGRESS SUMMARY

The project has moved beyond initial setup: authentication, database connectivity, health-record management, history, and the first analytics features are working. Htrack26 remains under active development, with clinical and research functionality forming the next major stages.

REFERENCES

[1] Flutter Documentation, https://docs.flutter.dev/. [2] Supabase Documentation, https://supabase.com/docs/. [3] fl_chart, https://pub.dev/packages/fl_chart/.